Quantitative Comparisons

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.

$\frac{7}{12} \approx 0.5833$, which is greater than 0.58. $A > B$.

2 [Foundational] • Target: Numeric comparisons

Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

$A = 56$. $B = 54 + 2 = 56$. $A = B$.

3 [Foundational] • Target: Numeric comparisons

Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.

$A = 579$. $B = 580$. $B > A$.

4 [Foundational] • Target: Numeric comparisons

Column A: 9^2. Column B: $8^2 + 15$. Compare the two columns.

$A = 81$. $B = 64 + 15 = 79$. $A > B$.

5 [Foundational] • Target: Numeric comparisons

Column A: $(-3)^2$. Column B: -3^2. Compare the two columns.

$A = (-3) \times (-3) = 9$. $B = -(3^2) = -9$ (no parentheses around the base). $A > B$.

6 [Foundational] • Target: Fraction and decimal comparisons

Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.

$\frac{3}{7} \approx 0.4286$, which is less than 0.43. $B > A$.

7 [Foundational] • Target: Fraction and decimal comparisons

Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.

$\frac{7}{12} \approx 0.5833$, which is greater than 0.58. $A > B$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Fraction and decimal comparisons

Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.

$2\frac{3}{8} = \frac{19}{8}$. $A = B$.

9 [Intermediate] • Target: Fraction and decimal comparisons

Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.

$\frac{5}{9} \approx 0.5556$, which is less than 0.6. $A > B$.

10 [Intermediate] • Target: Percent comparisons

Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

$A = 0.25 \times 160 = 40$. $B = 0.20 \times 200 = 40$. $A = B$.

11 [Intermediate] • Target: Percent comparisons

Column A: 30% of 90. Column B: 40% of 60. Compare the two columns.

$A = 0.3 \times 90 = 27$. $B = 0.4 \times 60 = 24$. $A > B$.

12 [Intermediate] • Target: Percent comparisons

Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

$A = 100 \times 1.1 \times 1.1 = 121$. $B = 100 \times 1.2 = 120$. $A > B$.

13 [Intermediate] • Target: Percent comparisons

Column A: 15% of 80. Column B: 12% of 100. Compare the two columns.

$A = 0.15 \times 80 = 12$. $B = 0.12 \times 100 = 12$. $A = B$.

14 [Intermediate] • Target: Exponent and root comparisons

Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.

$\sqrt{50} \approx 7.07$, which is less than 7.1. $B > A$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Exponent and root comparisons • Cross-Lesson: 3.01 Algebra Essentials

Column A: 2^{10}. Column B: 10^3. Compare the two columns.

$A = 1{,}024$. $B = 1{,}000$. $A > B$.

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16 [Challenging] • Target: Exponent and root comparisons • Cross-Lesson: 4.01 Geometry & Measurement
Column A: $\sqrt{0.09}$. Column B: 0.3. Compare the two columns.

$\sqrt{0.09} = 0.3$. $A = B$.

17 [Challenging] • Target: Exponent and root comparisons • Cross-Lesson: 6.01 Sequences & Patterns
Column A: 3^4. Column B: 4^3. Compare the two columns.

$A = 81$. $B = 64$. $A > B$.

18 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 7.01 Data Analysis & Probability
Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

$A = 2(5) + 3 = 13$. $B = 5 + 8 = 13$. $A = B$.

19 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 8.01 Word Problems & Reasoning
Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

$(x + 2)^2$ expands to $x^2 + 4x + 4$ for every value of x, so the columns are always equal. $A = B$.

20 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 1.01 Arithmetic & Number Sense
Given $n = -2$. Column A: n^2. Column B: n^3. Compare the two columns.

$A = (-2)^2 = 4$. $B = (-2)^3 = -8$. $A > B$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Geometry comparisons
Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater ✓**

$C = 2(3.14)(5) = 31.4$, which is less than 32.

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22 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column B is greater

d. **Column A is greater ✓**

Rectangle area $= 9 \times 4 = 36$; square perimeter $= 4 \times 7 = 28$. Column A is greater.

23 [Foundational] • Target: MCQ — Geometry comparisons

Column A: the hypotenuse of the right triangle shown. Column B: 16.

a. The relationship cannot be determined from the given information

b. **Column B is greater ✓**

c. The two columns are equal

d. Column A is greater

$\sqrt{9^2 + 12^2} = \sqrt{225} = 15$, which is less than 16.

24 [Foundational] • Target: MCQ — More geometry comparisons

Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. **Column B is greater ✓**

d. The two columns are equal

Cube $= 5^3 = 125$; box $= 4 \times 5 \times 7 = 140$. Column B is greater.

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25 [Foundational] • Target: MCQ — More geometry comparisons

Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.

a. Column B is greater

b. The two columns are equal

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Octagon $= (8 - 2) \times 180 = 1{,}080^\circ$; four triangles $= 4 \times 180 = 720^\circ$. Column A is greater.

26 [Foundational] • Target: MCQ — Numeric comparisons

Column A: 6×7. Column B: $5 \times 9 - 3$. Compare the two columns.

a. **The two columns are equal ✓**

b. Column B is greater

c. The relationship cannot be determined from the given information

d. Column A is greater

42 vs $45 - 3 = 42$: equal.

27 [Foundational] • Target: MCQ — Fraction and decimal comparisons

Column A: $\frac{5}{8}$. Column B: 0.62. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. **Column A is greater ✓**

$\frac{5}{8} = 0.625$, which is greater than 0.62.

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28 [Foundational] • Target: MCQ — Percent comparisons

Column A: 45% of 80. Column B: 40% of 100. Compare the two columns.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. **Column B is greater ✓**

36 vs 40: Column B is greater.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Exponent and root comparisons

Column A: 3^5. Column B: 5^3. Compare the two columns.

a. **Column A is greater ✓**

b. The two columns are equal

c. Column B is greater

d. The relationship cannot be determined from the given information

243 vs 125: Column A is greater.

30 [Intermediate] • Target: MCQ — Algebraic comparisons at a fixed value

Given $n = -3$. Column A: n^2. Column B: n^3. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. **Column A is greater ✓**

9 vs -27: Column A is greater.

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31 [Intermediate] • Target: MCQ — Comparisons requiring testing values

For any real number x: Column A: $(x+2)^2$. Column B: x^2+4x+4. Compare the two columns.

a. The two columns are equal ✓

b. Column A is greater

c. The relationship cannot be determined from the given information

d. Column B is greater

$(x+2)^2 = x^2+4x+4$ for every x: equal.

32 [Intermediate] • Target: MCQ — Geometry comparisons

Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater ✓

$V = 3.14 \times 9 \times 7 = 197.82$, which is less than 200.

33 [Intermediate] • Target: MCQ — Data and probability comparisons

A data set is $5, 8, 8, 10, 14$. Column A: the mean. Column B: the median.

a. The two columns are equal

b. Column B is greater

c. Column A is greater ✓

d. The relationship cannot be determined from the given information

Mean $= \frac{45}{5} = 9$; median $= 8$. Column A is greater.

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34 [Intermediate] • Target: MCQ — Word-problem comparisons

Layla's age is 4 years less than three times Huda's age. Huda is 9. Column A: Layla's age. Column B: 25.

a. Column A is greater

b. The two columns are equal

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

Layla $= 3(9) - 4 = 23$, which is less than 25.

35 [Intermediate] • Target: MCQ — Comparisons with inequalities and ranges

Given that $x < -4$. Column A: x. Column B: -4.

a. Column A is greater

b. **Column B is greater ✓**

c. The two columns are equal

d. The relationship cannot be determined from the given information

Every value less than -4 is smaller than -4, so Column B is always greater.

36 [Intermediate] • Target: MCQ — Exponent and root comparisons

Column A: 8×5. Column B: $7 \times 8 - 15$. Compare the two columns.

a. Column A is greater

b. The two columns are equal

c. The relationship cannot be determined from the given information

d. **Column B is greater ✓**

40 vs $56 - 15 = 41$: Column B is greater.

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37 [Intermediate] • Target: MCQ — Algebraic comparisons at a fixed value

Column A: 7×11. Column B: $11 \times 8 + 9$. Compare the two columns.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **Column B is greater ✓**

77 vs $88 + 9 = 97$: Column B is greater.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Comparisons requiring testing values • Cross-Lesson: 7.01 Data Analysis & Probability

Column A: 7×6. Column B: $4 \times 3 + 8$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. **Column A is greater ✓**

c. The two columns are equal

d. Column B is greater

42 vs $12 + 8 = 20$: Column A is greater.

39 [Challenging] • Target: MCQ — Geometry comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: 10% of 160. Column B: 20% of 40. Compare the two columns.

a. The two columns are equal

b. **Column A is greater ✓**

c. The relationship cannot be determined from the given information

d. Column B is greater

$A = 0.1 \times 160 = 16$. $B = 0.2 \times 40 = 8$. Column A is greater.

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40 [Challenging] • Target: MCQ — Ratio and proportion comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

Column A: 15% of 80. Column B: 20% of 290. Compare the two columns.

a. Column A is greater

b. The two columns are equal

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

$A = 0.15 \times 80 = 12$. $B = 0.2 \times 290 = 58$. Column B is greater.

41 [Challenging] • Target: MCQ — Data and probability comparisons • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Column A: 50% of 190. Column B: 30% of 70. Compare the two columns.

a. The relationship cannot be determined from the given information

b. **Column A is greater ✓**

c. The two columns are equal

d. Column B is greater

$A = 0.5 \times 190 = 95$. $B = 0.3 \times 70 = 21$. Column A is greater.

42 [Challenging] • Target: MCQ — Word-problem comparisons • Cross-Lesson: 3.01 Algebra Essentials

Column A: 30% of 210. Column B: 30% of 270. Compare the two columns.

a. Column A is greater

b. The two columns are equal

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

$A = 0.3 \times 210 = 63$. $B = 0.3 \times 270 = 81$. Column B is greater.

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43 [Challenging] • Target: MCQ — Comparisons with inequalities and ranges • Cross-Lesson: 4.01 Geometry & Measurement

Given $x = 8$. Column A: $2x + 5$. Column B: $2x + 9$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. **Column B is greater ✓**

d. The two columns are equal

$A = 2(8) + 5 = 21$. $B = 2(8) + 9 = 25$. Column B is greater.

44 [Challenging] • Target: MCQ — More geometry comparisons • Cross-Lesson: 6.01 Sequences & Patterns

For any real m. Column A: $|m|$. Column B: m. Compare the two columns.

a. Column B is greater

b. The two columns are equal

c. **The relationship cannot be determined from the given information ✓**

d. Column A is greater

If $m \geq 0$, the columns are equal; if $m < 0$, Column A is greater (since $|m| > 0 > m$) — the relationship is not constant.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Data Analysis & Probability

Column A: the perimeter of a rectangle with length 4 cm and width 3 cm. Column B: the perimeter of a square with side 14 cm.

a. **Column B is greater ✓**

b. The two columns are equal

c. Column A is greater

d. The relationship cannot be determined from the given information

$A = 2(4 + 3) = 14$ cm. $B = 4 \times 14 = 56$ cm. Column B is greater.

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